CS351 - IT Workshop II (Cloud Computing)

Assignment 8

Please follow the instructions below. Marks will be deducted if you do not follow these instructions.

(i) Screenshots: Take clear screenshots, resize the images to reduce their file size, and upload them to Google Classroom. Please do not upload a .zip file in Google Classroom.

(ii) Source Code Header: Include your name and roll number at the top of each source file using the following format.

Name: <Your Name>

Roll No: <Your Roll No>

(iii) Proper Comments: You need to write comments for each block of code, function inputs and outputs, and the working logic of the functions.

(iv) Proper Variable and Function Names: Please use meaningful variable and function names.

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1. In this assignment, we shall make use of the **Hadoop Streaming API** to write our **MapReduce** code. **Hadoop Streaming** uses UNIX standard streams as the interface between Hadoop and your program, so you can use any language that can read standard input and write to standard output to write your **MapReduce** program. However, we shall use **Python** for this assignment. All we need to do is write a script for the **mapper** and **reducer**, and **Hadoop** will take care of the rest. First, we shall simulate the behavior of a **MapReduce** program using simple UNIX commands to understand how the **mapper** and **reducer** work in Section 2.

2. **MapReduce Simulation:**

   For this simulation, we shall do a simple word count. The problem statement is: Given a text file (https://norvig.com/big.txt), get the count of every word in it. Now, the first task is to write the mapper.

Word Count Mapper

The job of the mapper is simple: read lines from stdin and spit out the key-value pairs to stdout. Write a Python script for the same. An example of expected output from the input is given in Table 1. Test your script by running cat inputfile | ./mapper.py

Input File	Mapper Output
hello world	hello,1
testing testing	world,1
hello	testing,1
	testing,1
	hello,1

Table 1: Mapper Example

Word Count Reducer

The next task is to write the script for the reducer. The reducer job is to take the output of the mapper and spit out the final count of every word to stdout. We will simulate the sorting phase of the Hadoop framework with the sort command, so that all the same words are ordered together. Test your script by cat inputfile | ./mapper.py | sort -t ','-k1 | ./reducer.py

Mapper Output	sort	Reducer Output
hello,1	hello,1	hello,2
world,1	hello,1	world,1
testing,1	testing,1	testing,2
testing,1	testing,1	
hello,1	world,1	

Table 2: Reducer Example

Run it in Hadoop

Now that we have written our **mapper** and **reducer**, we are ready to execute our program in **Hadoop**.

hadoop jar /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-*.jar -input path/to/inputfile -output path/to/outputdir -mapper path/to/mapper.py -reducer reducer.py

3. **Hadoop Assignment:** **Total Marks: 20**

- Implement a **map-reduce** program to find all distinct words in the file. Perform data cleaning in the mapper such that all punctuation is removed and all words become lowercase.

inputfile
Hello World!
Apache hadoop.
apache spark.

Map Reduce output
apache
hadoop
hello
spark
world

Table 3: Distinct Words MR example

- Extend the word count example to include a **combiner**. Simply use the -combiner combiner.py option.
- You are given a dataset of N points, and you are given the C candidate points. Implement a map-reduce program to assign each of the N points to the nearest (Euclidean distance) candidate point and update the candidate points by taking the average of all the points that were assigned to it. Make use of the iris dataset (http://archive.ics.uci.edu/ml/machine-learning-databases/iris/iris.data), which is in the format

  sepallength, sepalwidth, petallength, petalwidth, class

  Use the following three points given in Table 4 as candidate points.

  For this exercise, you may consider the sepal length and sepal width, and remove the rest.

5.8,4.0,1.2,0.2,Iris-setosa

6.1,2.8,4.0,1.3,Iris-versicolor

6.3,2.7,4.9,1.8,Iris-virginica

Table 4: Candidate Points